A fixed testing family on a bounded C1 source interval

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The source and the complete parent. For positive x, σ, z and $0 < b, h < 1$ put

$$r = bz < 1, \quad Q = 1 - b, \quad \theta = 1 - h, \quad e = xh/\sigma, \quad E_s = 1 - e, \quad K_s = 1 + x, \quad R_s = 1 + xr,$$
$$M_s = R_s - xz\theta, \quad J_s = 1 + xh - \sigma(1 - r),$$
$$P_s = \sigma + K_s + \sigma z(1 + bx + 2xh) + \sigma b x^2 h z^2 + x^2 h\theta z,$$
$$\pi = \frac{z(1 + xh)E_s + (1 - r)(1 + xhz)}{2P_s}, \qquad l = \frac{(1 + \sigma r)\{eR_s(1 + \sigma z) + M_s E_s\}}{2P_s},$$
$$\kappa_s = \frac{K_s(1 + \sigma r)(1 + \sigma z)}{2M_s P_s}.$$

The strict gaps are $E_s, M_s, J_s > 0$. The coefficients are

$$m = \kappa_s x^2 z h/\sigma, \quad C = \frac{K_s z(1 + \sigma r)E_s}{2(1 - r)P_s}, \quad A = \frac{\pi}{bz}, \quad \alpha = \frac{l}{Qz},$$
$$D = \frac{K_s(1 + \sigma z)(1 + \sigma r) - J_s P_s}{2J_s P_s}, \quad \beta = \frac{K_s(1 + \sigma r)(1 + xhz) - \theta P_s}{2\theta P_s}.$$

Require $D, \beta \geq 0$. With $D_u = 1 + u^2$, $D_v = 1 + v^2$, write the complete quarter-parent as

$$\begin{aligned}\mathcal{F} = {} & m(vW - X + uvY - uZ)^2 + CD_v(uW - Y)^2 + AD_u D_v Y^2 \\ & + (D + \tfrac{1}{2})(uW + uvX - Y - vZ)^2 + (\alpha + \tfrac{1}{2})D_u(vY - Z)^2 \\ & + (\beta + \tfrac{1}{2})D_v(X + uZ)^2 - \tfrac{1}{2}D_u D_v Z^2.\end{aligned}$$

The physical normalization is the pure testing factor $(D_u D_v)^{-1}$. Homogeneous polynomial values define both projective angle endpoints.

Theorem 1. *Fix rationals $0 < \ell \leq U$. Define the following rational constants:*

$$R = \max(1, 1/\ell), \quad \varepsilon_R = 2^{-40}(1 + R)^{-8}, \quad \delta_R = 2^{-20}(1 + R)^{-2},$$
$$\mu = \frac{1}{4\max(U, 1)}, \quad M = \lceil 96\max(U, 1)\rceil, \quad \tau = \delta_R^2/2,$$
$$\mathcal{B} = U^2(4 + 2M^2 + M^4) + U(2 + 2M^2 + M^4) + 1 + 2M^2,$$
$$k = \frac{\min(\ell, 1)\tau^2}{8\mathcal{B}}, \quad K = 3M^2, \quad \varepsilon = \min\left(2^{-78}, \varepsilon_R, \frac{k}{50(k + K)}\right).$$

Suppose the above native source satisfies

$$\ell \leq L = 1/z \leq U, \qquad 0 < x, \sigma \leq \frac{\varepsilon}{1 + z}.$$

There is one finite source-independent family of semialgebraic testing functions whose source Gram assignments express the physical normalization of the complete parent as a separated sum of squares at every pair of projective angles and every state. This statement includes $D = 0$, $\beta = 0$, every allowed source-opening rate, and all testing and state-coordinate zeros. Its inner-corner input is the bounded-R admission theorem on $\sqrt{|u|}, |w| \le \delta_R$, $w = 1/v$, for native sources $x, \sigma \le \varepsilon_R$, $0 < z \le R$. That theorem supplies fixed families of at most 220 and 157 functions in the two signs of its source angular coefficient.

All functions below are fixed after ℓ, U are chosen. They do not depend on L, r, h, any event coefficient, or a source residue.

Quantitative native admission. Put $H_0 = 1 + zQ$, $B_s = P_s + xz(J_s - \sigma)$ and

$$\mathcal{H} = R_s H_0 + z\{(1-r)xh(2+xr) - e[2 + x(1+h+r\theta)] + K_s e^2\}.$$

Direct substitution in the displayed source formulas gives

$$2\theta P_s \beta = hB_s - \sigma H_0, \qquad 2P_s(J_s D + x\theta\beta) = \sigma^2 \mathcal{H},$$

and hence

$$eB_s = xH_0 + \sigma\mathcal{H} - 2P_s J_s D/\sigma.$$

Writing $T = 1 + z$ and using $Tx, T\sigma \le \varepsilon$, $e < 1$ and $r < 1$ gives

$$1 \le P_s \le 1 + 4\varepsilon, \quad |M_s - 1| \le \varepsilon, \quad B_s \ge 1, \quad \mathcal{H} \le 4T, \quad e \le 5\varepsilon.$$

For

$$C_0 = \frac{1}{2L(1-r)}, \quad A_0 = \frac{1 + L(1-r)}{2Lr}, \quad a_0 = \frac{L}{2(1-Lr)},$$

the exact source ratios are

$$\frac{C}{C_0} = \frac{K_s(1+\sigma r)(1-e)}{P_s}, \qquad \frac{A}{A_0} = \frac{z(1+xh)(1-e) + (1-r)(1+xhz)}{P_s H_0},$$
$$\frac{\alpha}{a_0} = \frac{(1+\sigma r)\{eR_s(1+\sigma z) + M_s(1-e)\}}{P_s}.$$

The first two ratios lie between $1 - 9\varepsilon$ and $1 + 3\varepsilon$; the third lies between $1 - 5\varepsilon$ and $1 + 5\varepsilon$. For the second numerator use $H_0 - ze + xhz(2 - r - e)$ and $H_0 - z = 1 - r > 0$. For the third use its convex combination of M_s and $R_s(1 + \sigma z)$. In particular

$$d_s := |C/C_0 - 1| + |A/A_0 - 1| + |\alpha/a_0 - 1| \le 25\varepsilon \le \frac{k}{2(k+K)}, \qquad C, A \ge \mu. \tag{1}$$

No bound on h/σ and no inverse of D or β was used.

Lemma 2 (Both Bob charts at the Alice endpoint)**.** *If $C, A \ge \mu$ with $0 < \mu \le 1/4$, one fixed algebraic testing family suffices whenever $0 \le p = 1/u \le \mu/24$, for every Bob angle. The other nonnegative event coefficients are unrestricted.*

Proof. On $|v| \le 1$ use raw roots (W, Y, B, R) with $B = pX + Z$, $R = vX$. The complete homogeneous core is

$$K_p = B^2 - 2pvBW - 2vBY + (1+p^2)R^2 + 2(1+p^2)RW + (W - pY)^2 + (1+p^2)v^2Y^2.$$

Select only the fixed coefficients $C = A = \mu$. Their quarter-parent

$$Q_p = \mu(1+v^2)(W-pY)^2 + \mu(1+p^2)(1+v^2)Y^2 + K_p/2$$

has the testing-only matrix

$$\begin{pmatrix} \mu D_v + 1/2 & -p(\mu D_v + 1/2) & -pv/2 & (1+p^2)/2 \\ -p(\mu D_v + 1/2) & \mu D_v(1+2p^2) + (p^2 + (1+p^2)v^2)/2 & -v/2 & 0 \\ -pv/2 & -v/2 & 1/2 & 0 \\ (1+p^2)/2 & 0 & 0 & (1+p^2)/2 \end{pmatrix}.$$

At $p = 0$ its form is

$$\mu D_v(W^2+Y^2) + \tfrac{1}{2}(W+R)^2 + \tfrac{1}{2}(B-vY)^2.$$

The raw norm is at most three times the squared norm of $(W, Y, W+R, B-vY)$, so this matrix is at least $(\mu/3)I_4$. The difference from its $p = 0$ matrix has absolute row sums at most $3p$ for $|v| \le 1$, $p \le 1$, $\mu \le 1/4$.

On $|w| = |1/v| \le 1$ use raw roots (W, Y, X, H), $H = wZ$. The complete twice-homogeneous core is

$$\begin{aligned} K_{pw} = {} & H^2 - 2pHW + 2pwHX - 2HY + w^2W^2 + 2wWX - 2pw^2WY \\ & + (1 + p^2 + p^2w^2)(X^2+Y^2) - 2pwXY. \end{aligned}$$

The corresponding selected quarter-parent is $Q_{pw} = \mu D_w(W-pY)^2 + \mu(1+p^2)D_wY^2 + K_{pw}/2$. Its testing-only matrix is

$$\begin{pmatrix} \mu D_w + w^2/2 & -p(\mu D_w + w^2/2) & w/2 & -p/2 \\ -p(\mu D_w + w^2/2) & \mu D_w(1+2p^2) + (1+p^2D_w)/2 & -pw/2 & -1/2 \\ w/2 & -pw/2 & (1+p^2D_w)/2 & pw/2 \\ -p/2 & -1/2 & pw/2 & 1/2 \end{pmatrix}.$$

At $p = 0$ its form is

$$\mu D_w(W^2+Y^2) + \tfrac{1}{2}(X+wW)^2 + \tfrac{1}{2}(H-Y)^2 \succeq (\mu/3)I_4.$$

Its finite-p correction has absolute row sums at most $4p$. Thus both displayed matrices are at least $(\mu/6)I_4$ when $p \le \mu/24$. Factoring them takes place entirely in the testing field: they contain only fixed μ and testing variables. Their four Cholesky roots in each chart are therefore fixed algebraic testing functions. Every leading principal minor is strictly positive, so the usual finite Cholesky formulas define them at all chart points.

Restore the six original nonnegative contributions with source coefficients $C-\mu, A-\mu, \alpha, m, D, \beta$. On the v chart their fixed testing roots are

$$\begin{gathered} \sqrt{D_v}(W-pY), \quad \sqrt{(1+p^2)D_v}\,Y, \quad \sqrt{1+p^2}(vY-Z), \\ pvW - pX + vY - Z, \quad W + vX - pY - pvZ, \quad \sqrt{D_v}(pX+Z). \end{gathered}$$

On the w chart they are

$$\begin{gathered} \sqrt{D_w}(W-pY), \quad \sqrt{(1+p^2)D_w}\,Y, \quad \sqrt{1+p^2}(Y-wZ), \\ pW - pwX + Y - wZ, \quad wW + X - pwY - pZ, \quad \sqrt{D_w}(pX+Z). \end{gathered}$$

This is at most ten fixed functions per chart, including their four testing Cholesky roots. All original terms and literal endpoints are included. The physical normalizers are pure testing. □

A finite annulus for all remaining bounded Alice angles. Put $c = (1+v^2)^{-1/2}$, $s = vc$, $P = uW$, $T = cZ$. The selected compact-Bob quarter-parent is

$$S_0 = C_0(P-Y)^2 + A_0(1+u^2)Y^2 + a_0(1+u^2)(sY-T)^2 + K_B/2,$$

where

$$K_B = X^2 + c^2(P-Y)^2 + s^2Y^2 - 2sPT + 2ucX(sP - sY + T) + u^2s^2X^2 + u^2(sY-T)^2.$$

For its raw Gram G_0 in (P, X, Y, T), direct cofactors give

$$\operatorname{tr} G_0^{-1} = \frac{2(L^2N_2 + LN_1 + N_0)}{(1+t)\{Lq + t(1-q)\}\{1 + L(1-r)\}}, \quad t = u^2,\ q = c^2,$$

with

$$\begin{aligned} N_2 &= 1 + r^2 + (1-r)q\{1 + r(1-q)\} + (1-r)(2-qr)t + (1-r)^2(1-q)t^2, \\ N_1 &= 2(1-r)q + \{2 - rq(3-q)\}t + (1-r)(1-q)t^2, \\ N_0 &= 1 + (2-q)t. \end{aligned}$$

The strict $T = 0$ block and positive determinant establish $G_0 \succ 0$ unless $u = c = 0$. For $|u| \le M$,

$$N_2 \le 4 + 2M^2 + M^4, \quad N_1 \le 2 + 2M^2 + M^4, \quad N_0 \le 1 + 2M^2.$$

Since $L \ge \ell$,

$$(1+t)\{Lq + t(1-q)\} \ge \min(\ell, 1)(q+t).$$

Consequently $G_0 \succeq \min(\ell,1)\tau^2 I/(2\mathcal{B})$ on $u^2 + c^2 \ge \tau^2$, and is at least kI on the larger annulus $u^2 + c^2 \ge \tau^2/2$. The complete core matrices, both in raw coordinates and in $e = (P - Y, Y, sY - T, X)$, have absolute row sums at most $K = 3M^2$. Using the common source factor $1 - d_s$ and retaining all three event excesses, (1) therefore gives an actual raw reserve $k/2$ on that larger annulus.

Here is an explicit finite construction in separated fields. Define

$$N = K + 1 + 24K^2/k, \qquad h_* = \min(1, k/24), \qquad B_* = N + 1.$$

Clip C, A, α at the fixed real N and retain their excesses. Both the event-coordinate map and its inverse have squared Frobenius norm at most six. Thus the unmodified event-coordinate Gram is at least $k/12$. Its core has norm at most K. Splitting coordinates according to their source coefficient being above N, the unchanged small block is at least $k/12$, the clipped large block is at least $N - K$, and the off-diagonal norm is at most K. The exact inequality

$$2Kab \le (24K^2/k)a^2 + (k/24)b^2$$

proves a clipped reserve h_*I.

The clipped event-coordinate Gram is affine in the seven testing monomials

$$\eta = (u^2, c^2, s^2, s, ucs, uc, u^2s^2).$$

Its coefficient matrices are exactly the following, with C', A', a' denoting the three clipped source weights:

$$\begin{aligned} G_{\text{const}} = \operatorname{diag}(C', A', a', 1/2), \qquad A_1 = \operatorname{diag}(0, A', a' + 1/2, 0), \quad A_2 = E_{11}/2, \\ A_3 = -(E_{12} + E_{21} + E_{22})/2, \quad A_4 = (E_{13} + E_{31} + E_{23} + E_{32})/2, \end{aligned}$$

$$A_5 = (E_{14} + E_{41} + E_{24} + E_{42})/2, \quad A_6 = -(E_{34} + E_{43})/2, \quad A_7 = E_{44}/2.$$

Each has norm at most B_*. Choose the fixed integer and grid

$$n = \left\lceil \max\left(\frac{8M^2}{\tau^2}, \frac{56M^2 B_*}{h_*} \right) \right\rceil, \quad \delta_g = 4M^2/n,$$

$$a_{jk} = \left(j/n, \frac{n^2 - k^2}{n^2 + k^2}, \frac{2kn}{n^2 + k^2} \right), \quad -Mn \le j \le Mn, \quad -n \le k \le n.$$

All centers are rational. The circle parametrization $c = (1-t^2)/(1+t^2)$, $s = 2t/(1+t^2)$, $-1 \le t \le 1$, has both coordinate derivatives bounded by two. Thus the nearest parameter grid point gives errors at most $1/n$ in c, s. Retain centers with $u_j^2 + c_k^2 \ge \tau^2/2$. Charts within $1/n$ of a center in each coordinate cover the target annulus. On each chart $b_i = \eta_i - \eta_i(a)$ satisfies $|b_i| \le \delta_g$: the derivative bounds for $u^2 s^2$ give $2M + 2M^2 \le 4M^2$, and those of the other six monomials are smaller. The exact identity

$$G' = G'(a) - 7\delta_g B_* I + \tfrac{1}{2} \sum_{i=1}^{7} \{ (B_* I + A_i)(\delta_g + b_i) + (B_* I - A_i)(\delta_g - b_i) \}$$

has positive source matrices, its first matrix being at least $h_* I/2$. It uses only the fixed testing functions e_j and $e_j \sqrt{\delta_g \pm b_i}$. Restore the fixed roots $\sqrt{1+u^2}Y$, $\sqrt{1+u^2}(sY - T)$ and the three full m, D, β events. At most 65 functions per chart express the complete parent. No selected source/test matrix was square-rooted in a mixed field.

Completing the testing cover. When $|u| \ge M$, Lemma 2 applies because $1/M \le \mu/24$. For $|u| \le M$ outside the displayed annulus, $|u|, c < \tau$. Thus

$$|u| < \delta_R^2/2, \qquad |w| = c/\sqrt{1-c^2} < 2\tau = \delta_R^2 \le \delta_R.$$

These inequalities imply the bounded-R inner testing conditions $\sqrt{|u|}, |w| \le \delta_R$. Its source box contains the present box, since $z \le 1/\ell \le R$ and $x, \sigma \le \varepsilon \le \varepsilon_R$.

These are finitely many fixed testing charts. To obtain one family on the whole testing domain, order the charts and assign a point to the first chart containing it. Multiply each chart function by the indicator of its resulting disjoint semialgebraic piece and extend it by zero elsewhere. The identities then sum literally; all source Gram assignments remain source-only. This introduces no additional functions per chart. The indicators and radicals have fixed semialgebraic graphs. Before this summation divide the compact-Bob functions by $\sqrt{D_u}$, the endpoint v-chart functions by $\sqrt{(1+p^2)D_v}$, and the endpoint w-chart functions by $\sqrt{(1+p^2)D_w}$. Use the corresponding physical normalizer for the inner family, with its constant factor $1/2$ if its target is the full parent $4\mathcal{F}$. All charts then represent the same quadratic function $\mathcal{F}/(D_u D_v)$, including its homogeneous projective values. The exact sign gauges are

$$(u, W, X, Y, Z) \longmapsto (-u, W, X, -Y, -Z), \qquad (v, W, X, Y, Z) \longmapsto (-v, W, -X, Y, -Z).$$

They preserve every square term and the signed complete core, so negative p requires no additional source statement. Real/imaginary splitting handles complex states. This proves Theorem 1.

Scope. The fixed testing family is obtained for each specified interval $[\ell, U]$. It is not uniform over all positive real residues of L without fixing such an interval, and does not treat $z = 0$ or other source charts. The bounded-R inner functions depend on the fixed R; this theorem does not identify them with the evaluated order-731 bounded-z construction. When $\ell \geq 1$, one may instead use the published local669 theorem, choose $\tau = 2^{-25}$, and omit ε_R from the minimum. Then the same proof requires no bounded-R input. This is a C1 source-domain result, not a global semidefinite lift of the quantum set.